AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII

ANNUAL EXAMINATIONS 2021

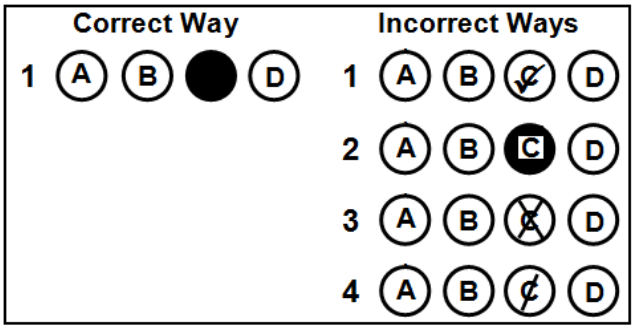
Mathematics

Time: 2 hours 10 minutes Marks: 65

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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 65 only.
4. In each question, there are four choices A, B, C, D. Choose ONE. On the answer grid, black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. The marks obtained on the 65 MCQs will be equated to the total marks of 100 for the theory examination results.
8. You may use a scientific calculator if you wish.

1. The domain of the function $f(x)=\dfrac{1}{(5-x)(5+x)}$ is

   A. R.
   B. $[-5,5]$.
   C. $(-5,5)$.
   D. $R-\{-5,5\}$.

2. If $f(x)=2-x$ and $g(x)=2+x$ are two functions, then the composition function $f\circ g$ will be

   A. 4
   B. x
   C. $-x$
   D. $4-x$

3. Which of the following functions is an even function?

   A. $f(x)=(x-2)^2$
   B. $f(x)=-x^2+3$
   C. $f(x)=x^2+\sin x$
   D. $f(x)=x-\cos x$

4. The value of $\lim\limits_{x\to 4}\dfrac{4}{\sqrt{x}-4}$ is

   A. -2
   B. 2
   C. 4
   D. undefined

5. The value of $\lim\limits_{x\to\infty}\dfrac{x-1}{x}$ is

   A. -1
   B. 0
   C. ∞
   D. 1

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6. For a function $f(x) = 3x^3 - 3$, the inverse function will be

A. $f^{-1}(x) = \sqrt[3]{\frac{x-3}{3}}$

B. $f^{-1}(x) = \sqrt[3]{\frac{x}{3}-3}$

C. $f^{-1}(x) = \sqrt[3]{\frac{x}{3}+3}$

D. $f^{-1}(x) = \sqrt[3]{\frac{x+3}{3}}$

7. At $y = 7$, the value(s) of x for $y = \sqrt{x^2 - 14x + 49}$ is/ are

A. -14
B. 14
C. 0 and 14
D. 0 and -14

8. The CORRECT representation of the interval $a \geq x \geq b$ is

A. (a, b)
B. $[a, b]$
C. (b, a)
D. $[b, a]$

9. It is given that $p = x$ and $q = x \sin x$. The derivative of $\frac{p}{q}$, with respect to x, is

A. $-\cos x$.
B. $\cos x$.
C. $\text{cosec}\, x \cot x$.
D. $-\text{cosec}\, x \cot x$.

10. If the rate of change of x and y with respect to time is 5 and 20 respectively, then the rate of change of y with respect to x is

A. $\frac{1}{4}$
B. 4
C. 25
D. 100

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11. If $k''(x) = x - x^3$, where $x \in [-1.5, 1.5]$, then $k(x)$ will be maximum at

A. $x = 0$
B. $x = \frac{1}{2}$
C. $x = \sqrt{2}$
D. $x = -\sqrt{2}$

12. The slope of tangent to a curve is given as $\frac{dy}{dx} = \sqrt{x} - 2$. If this tangent is parallel to $2y = x$ at a point, then the *x*-coordinate of this point on the curve is

A. 1
B. 9
C. $\frac{9}{4}$
D. $\frac{25}{4}$

13. The derivative of $a^2 - \frac{1}{x}$ with respect to *x* is

A. $\frac{1}{x^2}$.
B. $-\frac{1}{x^2}$.
C. $2a - \frac{1}{x^2}$.
D. $2a + \frac{1}{x^2}$.

14. If $y = \frac{e^x}{x}$, then $\frac{dy}{dx}$ will be

A. e^x.
B. e^{x-1}.
C. $\frac{e^x - xe^x}{x^2}$.
D. $\frac{xe^x - e^x}{x^2}$.

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15. If $xy = 0$, then $\frac{dy}{dx}$ will be

A. $-\frac{y}{x}$.

B. $-\frac{x}{y}$

C. $\frac{y}{x}$.

D. $\frac{x}{y}$.

16. If $y = \frac{\tan\theta}{\theta}$, then $\frac{dy}{d\theta}$ will be

A. $\frac{\theta\sec\theta - \tan\theta}{\theta^2}$.

B. $\frac{\tan\theta - \theta\sec\theta}{\theta^2}$.

C. $\frac{\tan\theta - \theta\sec^2\theta}{\theta^2}$.

D. $\frac{\theta\sec^2\theta - \tan\theta}{\theta^2}$.

17. If $y = \ln e^2$, then $\frac{dy}{dx}$ will be

A. -2

B. 0

C. 2

D. $\frac{2}{e}$

18. If $y = \ln e^{\tan x}$, then $\frac{dy}{dx}$ will be

A. $\sec^2 x$.

B. $\sec x \tan x$.

C. $\tan x \ln e^{\tan x - 1}$.

D. $\tan x \ln e^{\tan x - 1} \sec^2 x$.

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19. If $f(x) = \sin^2 x$, then second derivative of this function will be

A. $2\cos 2x.$
B. $-2\sin 2x.$
C. $2\sin 2x.$
D. $2\sin x\cos x.$

20. For the curve $y = 2x^3 - 1$, the slope of normal at the point $(-1,-3)$ will be

A. -6
B. $-\frac{1}{6}$
C. $\frac{1}{6}$
D. 6

21. For the function $f(x) = x$, where $f(-x) = -f(x)$, the value of $\int_{-\pi}^{\pi} x\,dx$ is

A. 0
B. $\frac{1}{2}$
C. π^2
D. $\frac{\pi^2}{2}$

Use the given information to answer Q.22 and Q.23.

It is given that $f(x) = (2 + x^2)^7$ and $k(x) = 2x.$

22. $\int f(x) \times k(x)dx$ is

A. $8(2 + x^2)^8 + C.$
B. $6(2 + x^2)^6 + C.$
C. $\frac{1}{6}(2 + x^2)^6 + C.$
D. $\frac{1}{8}(2 + x^2)^8 + C.$

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23. $\int \frac{k(x)}{f(x)} dx$ is

A. $(2+x^2)^{-6}+C.$

B. $(2+x^2)^{-8}+C.$

C. $-\frac{1}{6}(2+x^2)^{-6}+C.$

D. $-\frac{1}{8}(2+x^2)^{-8}+C.$

24. To convert $\frac{1}{(x^2+a)(x^2-b^2)}$ into its partial fractions, the appropriate form will be

A. $\frac{Ax+B}{(x^2+a)}+\frac{Cx+D}{x^2-b^2}.$

B. $\frac{Ax+B}{(x^2+a)}+\frac{C}{x^2-b^2}.$

C. $\frac{Ax+B}{(x^2+a)}+\frac{C}{x-b}+\frac{D}{x+b}.$

D. $\frac{Ax+B}{(x^2+a)}+\frac{C}{x-b}+\frac{D}{x^2-b^2}.$

25. The solution of the differential equation $x\cos y\frac{dy}{dx}=1$ is

A. $\sin y=\ln x+c.$

B. $-\sin y=\ln x+c.$

C. $\sin y=\frac{x^2}{2}+c.$

D. $-\sin y=\frac{x^2}{2}+c.$

26. The order and degree of the differential equation $5x\left(\frac{dy}{dx}\right)^3-\left(\frac{d^2y}{dx^2}\right)=y$, respectively are

A. 2 and 1
B. 2 and 3
C. 3 and 2
D. 3 and 1

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27. A stone, thrown vertically upward, reaches a height of $h = 30t + 5t^2$ after t seconds. The velocity of the stone is given by

A. $15t^2 + \frac{5}{3}t^3 \text{ ms}^{-1}$.

B. $60t^2 + 15t^3 \text{ ms}^{-1}$.

C. $30 + 10t \text{ ms}^{-1}$.

D. $t + 10t \text{ ms}^{-1}$.

28. The area under the curve $y = \sin x$ such that $-\pi \leq x \leq \pi$ can be expressed as

A. $\int_0^{2\pi} \sin x \, dx$.

B. $\int_{-\pi}^{\pi} \sin x \, dx$.

C. $\frac{d}{dx} \sin x \Big|_{\pi}^{2\pi}$.

D. $\frac{d}{dx} \sin x \Big|_{-\pi}^{0}$.

29. $\int \frac{dx}{\sqrt{a^2 - x}}$ is equal to

A. $\frac{1}{a} \sin^{-1} \frac{x}{a} + C$.

B. $2\sqrt{a^2 - x} + C$.

C. $-\sqrt{a^2 - x} + C$.

D. $-2\sqrt{a^2 - x} + C$.

30. $\int (1-x)^{\frac{1}{2}} \, dx$ is equal to

A. $\frac{3}{2}\sqrt{(1-x)^3} + C$.

B. $\frac{2}{3}\sqrt{(1-x)^3} + C$.

C. $-\frac{2}{3}\sqrt{(1-x)^3} + C$.

D. $-\frac{3}{2}\sqrt{(1-x)^3} + C$.

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31. $\int_{b}^{-b} dx$ is equal to

A. $-b$
B. $-2b$
C. 0
D. $2b$

32. $\int_{0}^{1} (x-1)dx$ is equal to

A. $-\frac{1}{2}$
B. $-\frac{1}{4}$
C. $\frac{1}{4}$
D. $\frac{1}{2}$

33. In normal form, $2x-y=1$ can be written as

A. $\frac{2x}{3}-\frac{y}{3}=\frac{1}{3}$
B. $\frac{2x}{5}-\frac{y}{5}=\frac{1}{5}$
C. $\frac{2x}{\sqrt{3}}-\frac{y}{\sqrt{3}}=\frac{1}{\sqrt{3}}$
D. $\frac{2x}{\sqrt{5}}-\frac{y}{\sqrt{5}}=\frac{1}{\sqrt{5}}$

34. The two parallel lines P and Q are given as $3y=6$ and $y=-8$. The perpendicular distance between P and Q is

A. 2 units.
B. 6 units.
C. 10 units.
D. 14 units.

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35. If the vertices $O(0,0)$, $P(0,a)$ and $Q(a,0)$ form a triangle, then the area of the triangle *OPQ* is

A. a^2 square units.
B. $\frac{a^2}{2}$ square units.
C. $2a^2$ square units.
D. $\frac{1}{2a^2}$ square units.

36. In Cartesian plane, the distance between the points $(0,-1)$ and $(\sqrt{3},0)$ is

A. $\sqrt{2}$ units.
B. 2 units.
C. 4 units.
D. $\sqrt{3}+1$ units.

37. In a Cartesian plane, two points $A\ (0,-1)$ and $B\ (-1,0)$ are given. If point $P(x, y)$ divides the line segment *AB* such that *AP:PB* = 2:1, then *x* is equal to

A. $-\frac{2}{3}$
B. $-\frac{1}{3}$
C. $\frac{1}{3}$
D. $\frac{2}{3}$

38. If the line $2y = x + 2$ is parallel to the line $y = -\frac{a}{b}x + c$, then $-\frac{a}{b}$ will be equal to

A. -1
B. $-\frac{1}{2}$
C. $\frac{1}{2}$
D. 1

39. The equation of a line which is parallel to the line $y = k$ and passes through the point (0, 0) will be

A. $x = 0$
B. $y = 0$
C. $y = k$
D. $x = k$

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40. The equation of a line which is perpendicular to the line $x = ky$ and passes through the point $(0, -1)$ will be

A. $y + 1 = kx.$
B. $y + 1 = \frac{1}{k}x.$
C. $y + 1 = -kx.$
D. $y + 1 = -\frac{1}{k}x.$

41. Which of the following points lies on the line $y = -2x - 2$?

A. $(0, 0).$
B. $(0, -2).$
C. $(1, 0).$
D. $(0, 1).$

42. If the lines $y = 2$, $x = b$ and $x = -y$ are concurrent, then the value of b will be

A. -2
B. -1
C. 1
D. 2

43. If the slopes of two coplanar lines l_1 and l_2 are $-a$ and $\frac{1}{a}$ respectively, then the angle from l_1 to l_2 will be

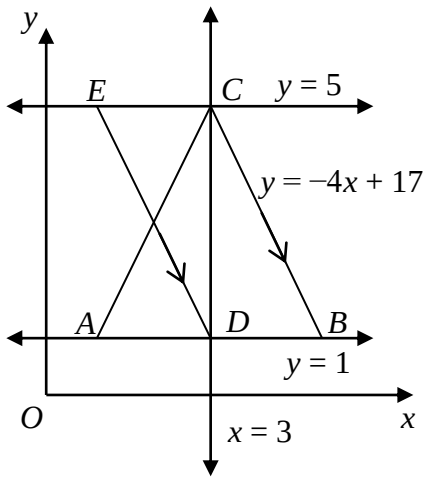
A. 0°
B. 90°
C. $\tan^{-1}\left(\frac{\frac{1}{a} + a}{2}\right)$
D. $\tan^{-1}\left(\frac{\frac{1}{a} + a}{1 - \frac{1}{a^2}}\right)$

44. If the line $y = 2$ is perpendicular to the line l, then the slope of line l will be

A. -2
B. $-\frac{1}{2}$
C. zero
D. undefined

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Use the given diagram to answer Q.45 and Q.46.



45. The x-intercept of the line CB is

A. -17
B. $\frac{17}{4}$
C. $-\frac{17}{4}$
D. 17

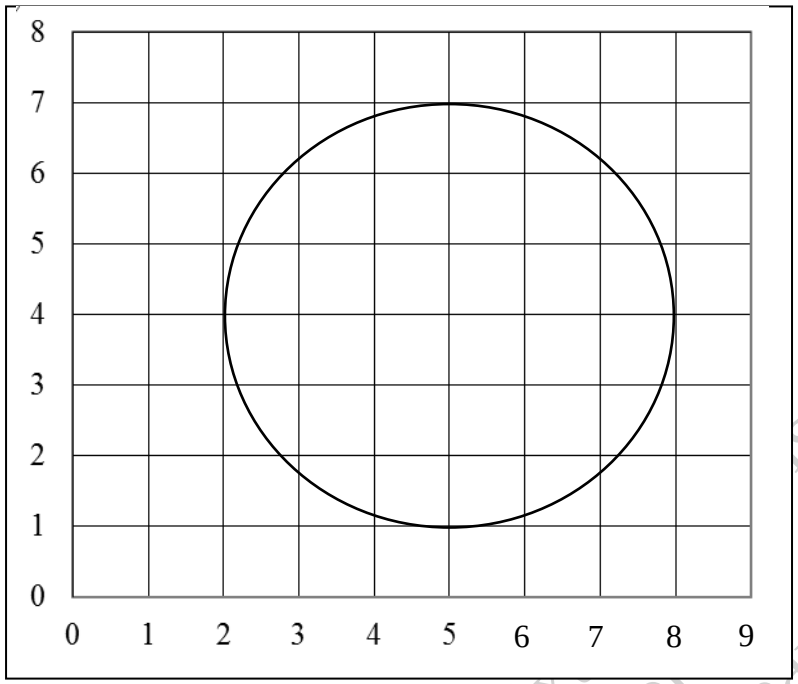
46. The coordinates of point B are

A. (1, 5).
B. (5, 1).
C. (4, 1).
D. (1, 4).

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Use the given information to answer Q.47 and Q.48.

The given figure shows a circle whose equation is $x^2 + y^2 + 2gx + 2fy + c = 0$.



47. If the value of c is 32, then the equation of circle is expressed as

A. $x^2 + y^2 - 10x - 8y + 32 = 0$
B. $x^2 + y^2 + 8x + 10y + 32 = 0$
C. $x^2 + y^2 + 10x + 8y + 32 = 0$
D. $x^2 + y^2 - 8x - 10y + 32 = 0$

48. One of the points that lies on this circle is

A. (5, 4).
B. (4, 4).
C. (7, 5).
D. (5, 7).

49. The radius of the circle represented by equation $(2x+3)^2 + (2y-4)^2 - 16 = 0$, is

A. 1
B. 2
C. 4
D. 16

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50. If the radius of circle $x^2 + y^2 - m = 0$ is 5 units, then the value of *m* will be

A. $\sqrt{5}$
B. 25
C. 10
D. 5

51. If the slope of the tangent to a circle at (a, b) is $\frac{m}{2}$, then the equation of this tangent will be

A. $y - b = mx - a.$
B. $y - b = m(x - a).$
C. $2(x - a) = m(y - b).$
D. $m(x - a) = 2(y - b).$

52. The radius of the circle given by the equation $x^2 + y^2 + 2x - 2 = 0$ is

A. 1
B. $\sqrt{2}$
C. $\sqrt{3}$
D. 2

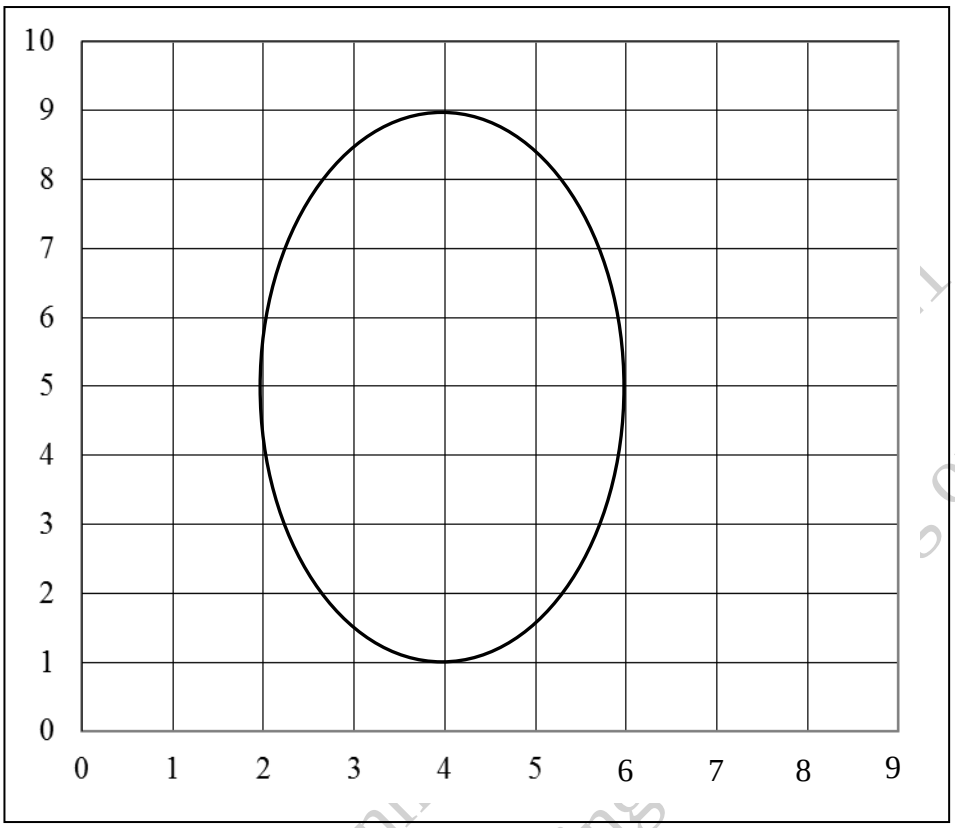
53. The length of the tangent from the point (0,0) to the circle $x^2 + y^2 + 2x + 2y + 8 = 0$ is

A. $2\sqrt{2}$
B. $\sqrt{10}$
C. 4
D. 16

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Use the given information to answer Q.54 and Q.55.

The ellipse shown in the graph can be represented by the equation $\frac{(x-h)^2}{b^2}+\frac{(y-k)^2}{a^2}=1.$



54. The co-vertices of this ellipse are

A. (4, 1), (4, 9).
B. (2, 5), (6, 5).
C. (0, 1), (0, 9).
D. (2, 0), (7, 0).

55. If the value of c is $2\sqrt{3}$, where $c=ae$, then the focus of the ellipse will be

A. $(\pm 2\sqrt{3}, 0)$.
B. $(0, \pm 2\sqrt{3})$.
C. $(4, 5+2\sqrt{3}), (4, 5-2\sqrt{3})$.
D. $(4+2\sqrt{3}, 5), (4-2\sqrt{3}, 5)$.

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56. For the ellipse $5(x-5)^2+25(y-5)^2=25$ the eccentricity is

A. $\frac{2}{\sqrt{5}}$

B. $\frac{\sqrt{5}}{2}$

C. $\frac{3}{\sqrt{5}}$

D. $\frac{\sqrt{5}}{3}$

57. Which of the following statements is/ are TRUE for the parabola $x^2=4ay$?

I. The opening of parabola is upward.
II. The curve lies in second and third quadrant.
III. The curve is symmetric with respect to x-axis.

A. I only
B. III only
C. I and II
D. II and III

58. The length of latus rectum of parabola $x^2=(-5)^2 y$ will be

A. 25
B. 10
C. $\frac{25}{4}$
D. $\frac{10}{4}$

59. If the eccentricity of a conic is greater than 1, then we have

A. $a^2=e^2-b^2$.
B. $b^2=a^2-e^2$.
C. $b^2=a^2(e^2-1)$.
D. $a^2=b^2(e^2-1)$.

60. The equation of ellipse is given by $\frac{x^2}{8^2}+\frac{y^2}{5^2}=1$. The ratio of the length of semi major axis to the length of semi minor axis is

A. 8:5
B. 5:8
C. 64:25
D. 25:64

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61. The equation of tangent to a hyperbola at a particular point is $mx+ny+k=0$. The slope of normal to the hyperbola at that point is

A. $-\frac{m}{n}$.

B. $-\frac{n}{m}$.

C. $\frac{n}{m}$.

D. $\frac{m}{n}$.

62. The equation of the directrix of the parabola $x^2+4y=0$ is

A. $x=1$
B. $y=1$
C. $x=-1$
D. $y=-1$

63. If the axis of a parabola is $y=1$ and vertex is (0, 1) and length of latus rectum is 1 unit, then the equation of parabola will be

A. $y=x^2-2x+1$

B. $y=x^2+2x+1$

C. $x=y^2+2y+1$

D. $x=y^2-2y+1$

64. The length of latera recta of the hyperbola $\frac{(x-2)^2}{1}-\frac{(y-1)^2}{4}=1$ is

A. $\frac{1}{4}$

B. $\frac{1}{2}$

C. 1

D. 8

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65. A hyperbola satisfies the given conditions.

I. centre at $(1,-1)$
II. length of semi conjugate axis is 3 units
III. length of semi transverse axis is 4 units
IV. conjugate axis is along x-axis

The equation of the hyperbola will be

A. $\dfrac{(y-1)^2}{9}-\dfrac{(x+1)^2}{16}=1$

B. $\dfrac{(y+1)^2}{16}-\dfrac{(x-1)^2}{9}=1$

C. $\dfrac{(x+1)^2}{16}-\dfrac{(y-1)^2}{9}=1$

D. $\dfrac{(x-1)^2}{9}-\dfrac{(y+1)^2}{16}=1$

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